

The Pitman-Yor Process and Pitman-Yor Process Mixture Models

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1 Introduction

The Pitman-Yor (PY) process generalizes the Dirichlet process (DP) by introducing a discount parameter d that modulates the probability of creating new clusters. Although both processes generate a discrete random measure $\tilde{P}$ that induces a partition of the latent parameters associated with the observations, their asymptotic properties concerning the number of clusters K_n differ significantly.

In the Dirichlet process, the expected number of clusters $\mathbb{E}[K_n]$ grows asymptotically at a logarithmic rate ($\mathbb{E}[K_n] \sim c \log(n)$, where $c > 0$ is the concentration parameter). This growth is often insufficient for modelling phenomena that require faster sublinear growth. The Pitman-Yor process addresses this limitation by allowing $\mathbb{E}[K_n]$ to grow asymptotically according to a power law ($\mathbb{E}[K_n] \propto n^d$, with $0 < d < 1$).

2 The Pitman-Yor process and its properties

Definition 2.1 (Pitman-Yor process). *Let $d \in [0, 1)$ be the discount parameter, $c > -d$ the concentration parameter, and G_0 a probability distribution on Θ , called the base probability measure. For $j = 1, 2, \dots$, let $Y_j \stackrel{\text{iid}}{\sim} G_0$ and $W_j \stackrel{\text{ind}}{\sim} \text{Beta}(1 - d, c + jd)$, with $p_1 = W_1$ and $p_j = W_j \prod_{r=1}^{j-1} (1 - W_r)$ for $j = 2, 3, \dots$. Assume that the sequences $(Y_j)_{j \geq 1}$ and $(p_j)_{j \geq 1}$ are independent. The random probability measure*

$$\tilde{P}(A) = \sum_{j=1}^{\infty} p_j \delta_{Y_j}(A), \quad \text{for every } A \in \mathcal{F},$$

defined on Θ , is called a Pitman-Yor process. We write $\tilde{P} \sim \text{PY}(d, c, G_0)$.

The Dirichlet process is the special case of the Pitman-Yor process with $d = 0$.

Proposition 2.1 (PY predictive distribution). *Consider an exchangeable sequence $(\theta_i)_{i \geq 1}$ of random variables on Θ such that $\theta_i \mid \tilde{P} \stackrel{\text{iid}}{\sim} \tilde{P}$, where $\tilde{P} \sim \text{PY}(d, c, G_0)$. The predictive distribution of θ_n , conditional on $\theta_1, \dots, \theta_{n-1}$, is*

$$P(\theta_n \in A \mid \theta_1, \dots, \theta_{n-1}) = \frac{c + k_{n-1}d}{c + n - 1} G_0(A) + \sum_{j=1}^{k_{n-1}} \frac{n_j - d}{c + n - 1} \delta_{\theta_j^*}(A),$$

for every measurable subset A of Θ . Here k_{n-1} is the number of clusters generated by the first $n - 1$ parameters, θ_j^* is the value associated with the j th cluster, and n_j its cardinality.

Translation note. The summation limit $k_n - 1$ is retained as printed in the original report; the accompanying definition refers to k_{n-1} .

The influence of d appears in sequential sampling: the probability that an existing cluster attracts a new observation is proportional to $n_j - d$. For a newly created cluster ($n_j = 1$), a large value of d significantly reduces the probability of further growth, generating many small clusters. However, as n_j increases, the relative effect of the discount diminishes. This mechanism produces a power-law distribution characterized by many small clusters and a few very large clusters.

Definition 2.2 (PY mixture model). *Given a Pitman-Yor process $\tilde{P} \sim \text{PY}(d, c, G_0)$, the corresponding hierarchical mixture model is defined by*

$$\begin{aligned} \theta_i &\mid \tilde{P} \stackrel{\text{iid}}{\sim} \tilde{P}, \\ X_i &\mid \theta_i \stackrel{\text{iid}}{\sim} K(x_i; \theta_i), \quad i = 1, \dots, n. \end{aligned}$$

Proposition 2.2 (Exchangeable partition probability function). *Given a Pitman-Yor process $\tilde{P} \sim \text{PY}(d, c, G_0)$, the prior probability of a particular partition of the parameters $\{\theta_1, \dots, \theta_n\}$ into k blocks of sizes $\{n_1, \dots, n_k\}$ has the form*

$$\Pi_k^{(n)}(n_1, \dots, n_k) = \frac{\prod_{j=1}^{k-1} (c + dj)}{(c + 1)_{n-1}} \prod_{j=1}^k (1 - d)_{n_j - 1}.$$

Proof. By exchangeability of the θ_i , we may consider an ordered partition

$$\begin{aligned} \Pi_k^{(n)} = & \left\{ \underbrace{\{\theta_1, \dots, \theta_{n_1}\}}_{\text{Block 1}}, \underbrace{\{\theta_{n_1+1}, \dots, \theta_{n_1+n_2}\}}_{\text{Block 2}}, \dots, \right. \\ & \left. \underbrace{\{\theta_{n_1+\dots+n_{k-1}+1}, \dots, \theta_{n_1+\dots+n_{k-1}+n_k}\}}_{\text{Block } k} \right\}. \end{aligned}$$

For the first block,

$$\begin{aligned}
P(\theta_1 = \dots = \theta_{n_1}) &= \int p(\theta_1 = x) p(\theta_2 = \theta_1 \mid \theta_1 = x) \dots \\
&\quad \cdot p(\theta_{n_1} = \theta_1 \mid \theta_1 = \dots = \theta_{n_1-1} = x) \, dx \\
&= \frac{1-d}{c+1} \frac{2-d}{c+2} \dots \frac{n_1-1-d}{c+n_1-1} \int g_0(x) \, dx \\
&= \frac{(1-d)_{n_1-1}}{(c+1)_{n_1-1}}.
\end{aligned}$$

Next, for the second block,

$$\begin{aligned}
P(\theta_{n_1+1} = \dots = \theta_{n_1+n_2} \neq \theta_1 \mid \theta_1 = \dots = \theta_{n_1}) \\
&= \int p(\theta_{n_1+1} = y \neq \theta_1 \mid \theta_1 = \dots = \theta_{n_1}) \dots \\
&\quad \cdot p(\theta_{n_1+n_2} = \theta_{n_1+1} \neq \theta_1 \mid \theta_1 = \dots = \theta_{n_1}, \\
&\quad \quad \quad \theta_{n_1+1} = \dots = \theta_{n_1+n_2-1} = y) \, dy \\
&= \frac{c+d}{c+n_1} \frac{1-d}{c+n_1+1} \dots \frac{n_2-1-d}{c+n_1+n_2-1} \int g_0(y) \, dy \\
&= (c+d) \frac{(1-d)_{n_2-1}}{(c+n_1)_{n_2}}.
\end{aligned}$$

Iterating over the k blocks and combining the terms gives

$$\begin{aligned}
\Pi_k^{(n)}(n_1, \dots, n_k) &= \frac{(1-d)_{n_1-1}}{(c+1)_{n_1-1}} (c+d) \frac{(1-d)_{n_2-1}}{(c+n_1)_{n_2}} \dots \\
&\quad \cdot [c + (k-1)d] \frac{(1-d)_{n_k-1}}{(c+n_1+\dots+n_{k-1})_{n_k}} \\
&= \frac{\prod_{j=1}^{k-1} (c+dj)}{(c+1)_{n-1}} \prod_{j=1}^k (1-d)_{n_j-1}.
\end{aligned}$$

□

3 Prior number and size of clusters

3.1 The Dirichlet process

Proposition 3.1 (Prior number of DP clusters). *Given a Dirichlet process with concentration parameter $c > 0$ and base measure G_0 , the expected prior number of clusters in a sample of size n is*

$$\mathbb{E}[K_n] = \sum_{i=1}^n \frac{c}{c+i-1}.$$

Proof. Define an indicator variable $\mathbf{1}_n$ for the event that the n th draw creates a new cluster:

$$\mathbf{1}_n = \begin{cases} 1 & \text{if a new cluster is generated,} \\ 0 & \text{otherwise.} \end{cases}$$

Using the predictive distribution to calculate the expectation,

$$\begin{aligned} \mathbb{E}[K_n] &= \mathbb{E}\left[\sum_{i=1}^n \mathbf{1}_i\right] = \sum_{i=1}^n \mathbb{E}[\mathbf{1}_i] = \sum_{i=1}^n P(\mathbf{1}_i = 1) \\ &= \sum_{i=1}^n \frac{c}{c + i - 1}. \end{aligned}$$

□

This result establishes that, although the process can generate an unlimited number of distinct clusters, the discovery rate of new clusters decreases hyperbolically with i . Asymptotically, as n tends to infinity, the expected number of clusters grows logarithmically with the number of observations n . Formally,

$$\mathbb{E}[K_n] \sim c \ln(n),$$

where c is the concentration parameter.

The parameter c acts as a prior on the number of clusters induced by the process. Although direct proportionality holds only asymptotically, c still acts as a scale factor: small values favour a high concentration of the data in a few groups, whereas large values encourage the discovery of more distinct clusters.

Proposition 3.2 (Prior DP cluster mass). *Given a Dirichlet process $\tilde{P} \sim \text{DP}(c, G_0)$, the expected mass of the j th cluster in the stick-breaking construction is*

$$\mathbb{E}[p_j] = \frac{1}{c + 1} \left(\frac{c}{c + 1} \right)^{j-1}.$$

Proof. From the stick-breaking definition of the j th mass,

$$p_j = W_j \prod_{r=1}^{j-1} (1 - W_r),$$

its expectation can be calculated using $W_j \stackrel{\text{iid}}{\sim} \text{Beta}(1, c)$ for $j = 1, 2, \dots$:

$$\begin{aligned} \mathbb{E}[p_j] &= \mathbb{E} \left[W_j \prod_{r=1}^{j-1} (1 - W_r) \right] \\ &= \mathbb{E}[W_j] \prod_{r=1}^{j-1} \mathbb{E}[1 - W_r] \\ &= \frac{1}{c+1} \left(\frac{c}{c+1} \right)^{j-1}. \end{aligned}$$

□

This geometric decay reflects the tendency of the process to concentrate mass on the first generated atoms. The concentration parameter controls the speed of this decay: small values of c accentuate concentration on few components, whereas large values spread mass over more atoms.

Unlike the stick-breaking ordering, for masses rearranged in decreasing order, where (j) denotes rank ($p_{(1)} \geq p_{(2)} \dots$), there is no simple closed-form analytical expression for the exact expectation. However, the asymptotic behaviour as $j \rightarrow \infty$ is known:

$$\mathbb{E}[p_{(j)}] \sim \mathcal{C}_{\text{DP}} e^{-j},$$

where $\mathcal{C}_{\text{DP}}$ is a constant that depends on the process parameters. *Translation note.* The expression above is retained from the source. Its displayed decay rate contains no c , and the reported comparison uses $c = 1$; it is not established here as a general rate for other values of c .

This behaviour highlights an exponential (very rapid) decay in cluster sizes, implying that the probability mass is concentrated mainly on the first ordered components (the largest clusters), leaving negligible weight for clusters in the right tail.

A Monte Carlo study is developed below to confirm the theoretical results empirically. In particular, the behaviour of clusters ordered by size is analysed through simulation and compared with the theoretical reference for the order of creation, to highlight their difference. The Pitman-Yor process was simulated using the predictive distribution.

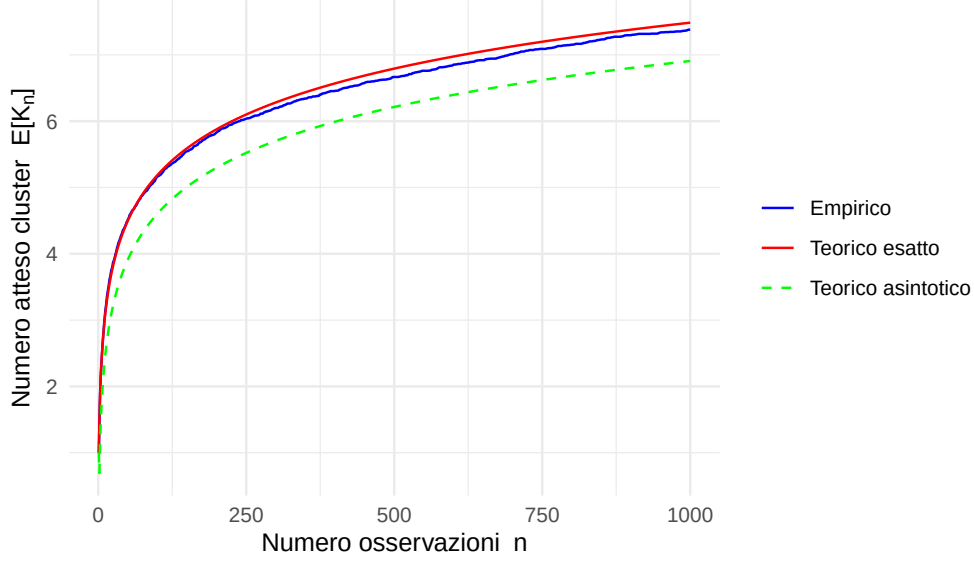


Figure 1: Expected number of clusters in the process $DP(c = 1)$: comparison between simulation and theoretical curves ($R = 300$ replications).

Original figure labels: “Numero osservazioni n ”: number of observations n ; “Numero atteso cluster $E[K_n]$ ”: expected number of clusters $E[K_n]$; “Empirico”: empirical; “Teorico esatto”: exact theoretical; “Teorico asintotico”: asymptotic theoretical.

As shown in Figure 1, the simulated mean number of clusters (blue curve) converges rapidly and almost perfectly to the exact expected number (red curve). This close correspondence confirms that 300 replications are sufficient to obtain a precise estimate of $\mathbb{E}[K_n]$, minimizing Monte Carlo variance.

The asymptotic approximation (dashed green line) provides an excellent estimate, progressively approaching the exact value as n increases. The simulation therefore validates that the Dirichlet process implements a parsimonious clustering mechanism, in which logarithmic growth of $\mathbb{E}[K_n]$ ensures that the model tends to reuse existing clusters.

Figure 2a shows the excellent overlap between the sample proportion (blue line) and the theoretical proportion (dashed red line). The plot clearly illustrates the rapid convergence to zero of the expected mass, confirming that in the stick-breaking process almost all the probability is concentrated on the very first generated clusters.

In Figure 2b, the empirical mean of masses ordered by rank lies systematically above the stick-breaking reference line (dashed red line) at the first value of j . This initial difference highlights a crucial phenomenon: the rank-1 cluster (the largest) captures, on average, a significantly greater share of mass than the first cluster generated in time. This confirms that, although the process favours the first generated components, the dominant cluster does not necessarily coincide with the first in order of appearance.

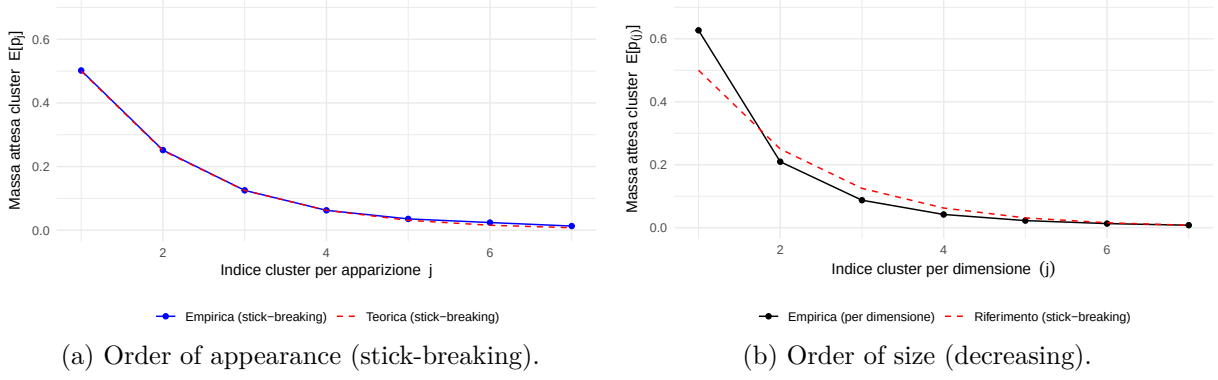


Figure 2: Expected cluster mass in the process $DP(c = 1)$: comparison between simulation and theoretical curves ($R = 300$ replications). In panel (b), the theoretical expected mass for the order of appearance (dashed red line) is shown as a reference to highlight the systematic difference caused by reordering by size.

Original figure labels: “Indice cluster per apparizione j ”: cluster index by order of appearance j ; “Indice cluster per dimensione (j)”: cluster index by size (j); “Massa attesa cluster”: expected cluster mass; “Empirica (stick-breaking)”: empirical (stick-breaking); “Teorica (stick-breaking)”: theoretical (stick-breaking); “Empirica (per dimensione)”: empirical (ordered by size); “Riferimento (stick-breaking)”: reference (stick-breaking).

For $j > 1$, the ordered empirical line tends to fall below the theoretical reference. This behaviour follows from the normalization constraint $\sum_{j=1}^{\infty} p_j = 1$: because the first ordered clusters absorb more probability mass than in the stick-breaking construction, the remaining mass available to lower-ranked clusters is necessarily smaller, causing more pronounced decay in the tail of the distribution.

3.2 The Pitman-Yor process

Proposition 3.3 (Prior number of PY clusters). *Given a Pitman-Yor process $\tilde{P} \sim PY(d, c, G_0)$, the expected prior number of clusters in a sample of size n is*

$$\mathbb{E}[K_n] = \frac{c}{d} \left\{ \prod_{i=1}^n \frac{c + d + i - 1}{c + i - 1} - 1 \right\}.$$

Proof. Define an indicator variable $\mathbf{1}_n$ for the event that the n th draw creates a new cluster:

$$\mathbf{1}_n = \begin{cases} 1 & \text{if a new cluster is generated,} \\ 0 & \text{otherwise.} \end{cases}$$

Calculate the expectation iteratively using the predictive distribution:

$$\begin{aligned}
\mathbb{E}[K_n] &= \mathbb{E}[K_{n-1}] + \mathbb{E}[\mathbf{1}_n] \\
&= \mathbb{E}[K_{n-1}] + \mathbb{E}\left[\frac{c + K_{n-1}d}{c + n - 1}\right] \\
&= \mathbb{E}[K_{n-1}] + \frac{c}{c + n - 1} + \frac{d}{c + n - 1}\mathbb{E}[K_{n-1}] \\
&= \mathbb{E}[K_{n-1}] \left(\frac{c + n - 1 + d}{c + n - 1}\right) + \frac{c}{c + n - 1}.
\end{aligned}$$

Since $\mathbb{E}[K_1] = 1$, the equation can be solved recursively to obtain

$$\mathbb{E}[K_n] = \frac{c}{d} \left\{ \prod_{i=1}^n \frac{c + d + i - 1}{c + i - 1} - 1 \right\}.$$

□

Although the exact expression is in closed form, it is difficult to interpret. However, rewriting the product in terms of Gamma functions and examining its behaviour as $n \rightarrow \infty$ reveals faster asymptotic growth than in the Dirichlet process:

$$\mathbb{E}[K_n] \sim S \cdot n^d, \quad S = \frac{\Gamma(c + 1)}{d\Gamma(c + d)}.$$

The expected number of clusters thus follows a power law in n . This asymptotic behaviour makes the Pitman-Yor process ideal for modelling phenomena in which the probability of discovering new clusters decreases much more slowly than predicted by the Dirichlet process.

Proposition 3.4 (Prior PY cluster mass). *Given a Pitman-Yor process $\tilde{P} \sim \text{PY}(d, c, G_0)$, the expected mass of the j th cluster in the stick-breaking construction is*

$$\mathbb{E}[p_j] = \frac{1 - d}{1 + c + (j - 1)d} \prod_{r=1}^{j-1} \frac{c + rd}{1 + c + (r - 1)d}.$$

Proof. From the stick-breaking construction for the j th mass,

$$p_j = W_j \prod_{r=1}^{j-1} (1 - W_r),$$

its expectation can be calculated by recalling that, for $j = 1, 2, \dots$, the random variables W_j are independent and distributed as $\text{Beta}(1 - d, c + jd)$. Hence,

$$\begin{aligned}\mathbb{E}[p_j] &= \mathbb{E}\left[W_j \prod_{r=1}^{j-1} (1 - W_r)\right] \\ &= \mathbb{E}[W_j] \prod_{r=1}^{j-1} \mathbb{E}[1 - W_r] \\ &= \frac{1 - d}{1 + c + (j - 1)d} \prod_{r=1}^{j-1} \frac{c + rd}{1 + c + (r - 1)d}.\end{aligned}$$

□

This expression generalizes the geometric decay typical of the Dirichlet process. Specifically, d reduces the expected breaking fraction W_j for each j : this makes the subdivision process more conservative, slowing the erosion of the available mass. This effect is reflected in much slower decay of the product of residual fractions, thus assigning non-negligible weight even to clusters with high indices.

As discussed for the Dirichlet process, masses reordered in decreasing order in the Pitman-Yor process also have no simple closed-form analytical expression for their exact expectation. However, the asymptotic behaviour as $j \rightarrow \infty$ is known:

$$\mathbb{E}[p_{(j)}] \sim \mathcal{C}_{\text{PY}} j^{-1/d},$$

where $\mathcal{C}_{\text{PY}}$ is a constant that depends on the process parameters.

In Figure 3, although linearity on a logarithmic scale confirms the general power-law behaviour, this tendency weakens for values of d close to zero. In this case, the distribution curves downwards, moving away from the theoretical straight line. This deviation is attributable to the asymptotic nature of the approximation (valid as $j \rightarrow \infty$), which loses accuracy for the first ranks. Conversely, as d increases, linearity becomes pronounced even for the first ranks and the slope decreases, indicating slower decay of the expected mass, which is spread over far more clusters than for small d .

This relation indicates that the expected size of a cluster is inversely proportional to a power of its rank j . This particular power-law behaviour constitutes the formal definition of Zipf's law. Zipf's law is an empirical law stating that, when a set of measured values is ordered in decreasing order, the value of the n th element is often approximately inversely proportional to n , which is called its rank.

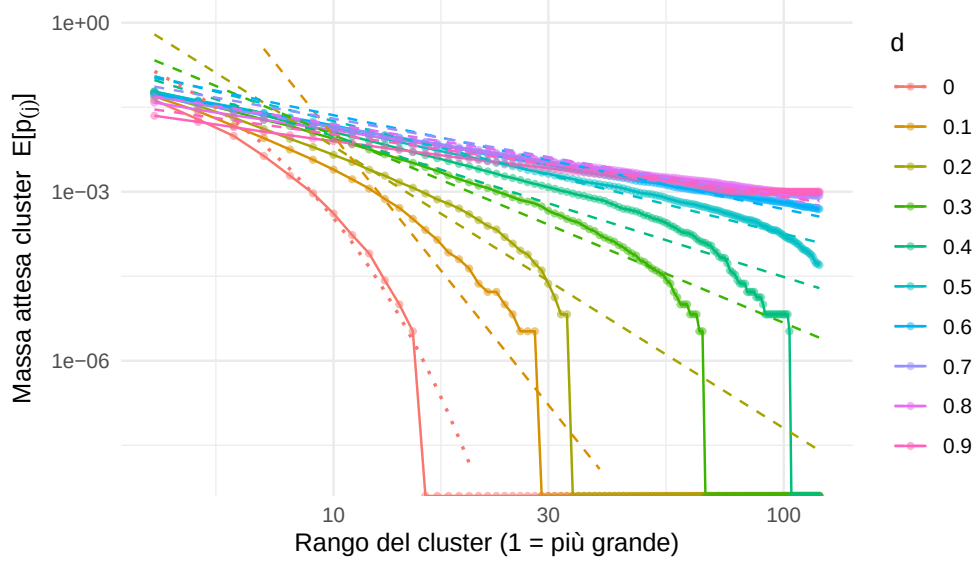


Figure 3: Rank distribution in the process $PY(c = 1, d)$ for varying d : comparison between simulation (solid lines) and asymptotic theoretical curves (dashed lines) on a logarithmic scale ($R = 300$ replications).

Original figure labels: “Rango del cluster (1 = più grande)”: cluster rank (1 = largest); “Massa attesa cluster $E[p_{(j)}]$ ”: expected cluster mass $E[p_{(j)}]$.

The best-known example of Zipf’s law concerns the frequency distribution of words in a text.

Unlike the exponential decay of the Dirichlet process, this polynomial behaviour is much slower and implies that a significant share of the mass remains distributed over the many high-ranked clusters in the right tail.

As before, the behaviour of clusters ordered by size is analysed through simulation and compared with the theoretical reference for the order of creation, to highlight their difference.

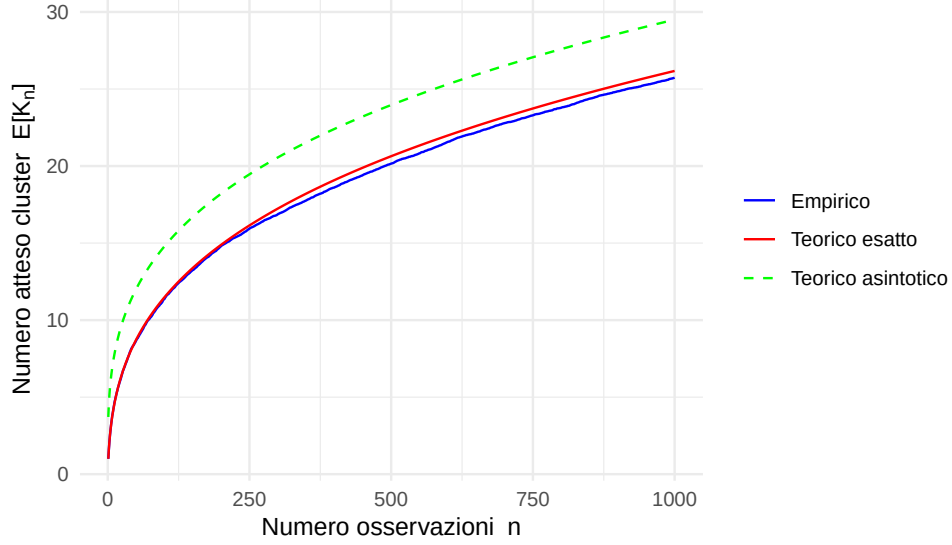


Figure 4: Expected number of clusters in the process $PY(d = 0.3, c = 1)$: comparison between simulation and theoretical curves ($R = 300$ replications).

Original figure labels: “Numero osservazioni n ”: number of observations n ; “Numero atteso cluster $E[K_n]$ ”: expected number of clusters $E[K_n]$; “Empirico”: empirical; “Teorico esatto”: exact theoretical; “Teorico asintotico”: asymptotic theoretical.

As can be seen in Figures 4 and 5, the discount parameter is responsible for the heavy-tail behaviour: increasing d accelerates the growth of the number of clusters and slows the decay of their expected mass, preserving probability for rare clusters. In contrast, the concentration parameter controls the general propensity to generate new categories: larger c increases the probability of creating a new cluster, leading to higher $\mathbb{E}[K_n]$ and a slight increase in the dispersion of mass. Thus d models the shape of the decay (acceleration or slowing), whereas c models its magnitude by acting as a scale factor.

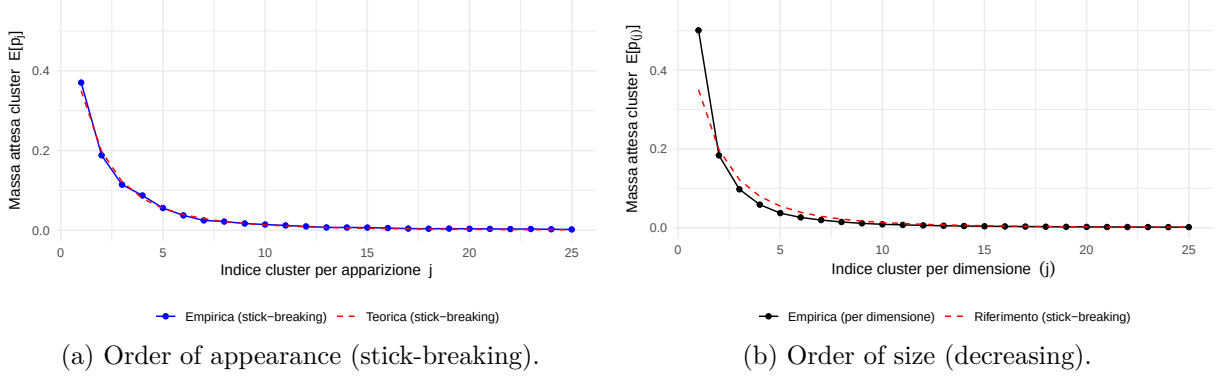


Figure 5: Expected cluster mass in the process $PY(d = 0.3, c = 1)$: comparison between simulation and theoretical curves ($R = 300$ replications). In panel (b), the theoretical expected mass for the order of appearance (dashed red line) is shown as a reference to highlight the systematic difference caused by reordering by size.

Original figure labels: “Indice cluster per apparizione j ”: cluster index by order of appearance j ; “Indice cluster per dimensione (j)”: cluster index by size (j); “Massa attesa cluster”: expected cluster mass; “Empirica (stick-breaking)”: empirical (stick-breaking); “Teorica (stick-breaking)”: theoretical (stick-breaking); “Empirica (per dimensione)”: empirical (ordered by size); “Riferimento (stick-breaking)”: reference (stick-breaking).

3.3 Expected number of clusters by size

Proposition 3.5. Let $\Pi^{(n)}$ be a random partition of the parameter set $\{\theta_1, \dots, \theta_n\}$ generated by a Pitman-Yor process $\tilde{P} \sim PY(d, c, G_0)$. Let $K_{n,r}$ denote the random variable counting the number of clusters in $\Pi^{(n)}$ with cardinality exactly r . Its expectation is

$$\mathbb{E}[K_{n,r}] = \binom{n}{r} \frac{(1-d)_{r-1}(c+d)_{n-r}}{(c+1)_{n-1}}.$$

Proof. By exchangeability of the sequence $\{\theta_i\}_{i=1}^n$, the probability that a subset of parameters forms a cluster depends only on its cardinality, not on its specific indices. We can therefore fix an arbitrary subset of size r , for example the first r elements $A^* = \{\theta_1, \dots, \theta_r\}$, and multiply the probability that it forms a cluster by the number of ways to select that subset:

$$\mathbb{E}[K_{n,r}] = \binom{n}{r} P(A^* \in \Pi^{(n)}).$$

Consider a configuration in which the first cluster has size $n_1 = r$, while the remaining $k-1$ clusters partition the remaining $n-r$ parameters. The probability of this configuration is given by the EPPF:

$$\Pi_k^{(n)}(n_1, \dots, n_k) = \frac{\prod_{j=1}^{k-1} (c+d_j)}{(c+1)_{n-1}} (1-d)_{r-1} \prod_{j=2}^k (1-d)_{n_j-1}.$$

In the first product, isolate the term for $j = 1$ and change the index to $l = j - 1$ in the remaining terms:

$$\begin{aligned}\prod_{j=1}^{k-1} (c + dj) &= (c + d) \prod_{j=2}^{k-1} (c + dj) \\ &= (c + d) \prod_{l=1}^{k-2} ((c + d) + dl).\end{aligned}$$

The resulting product has the coefficient structure of a process with shifted concentration parameter $c' = c + d$ and one fewer cluster. Substituting this decomposition, we calculate the marginal probability by summing over all possible partitions π' of the remaining $n - r$ elements (the set of partitions is denoted by $\mathcal{P}_{n-r}$). For each partition π' , write k' for its number of clusters and $m_1, \dots, m_{k'}$ for their sizes. The marginal probability becomes

$$\begin{aligned}P(A^* \in \Pi^{(n)}) &= \frac{(c + d)(1 - d)_{r-1}}{(c + 1)_{n-1}} \\ &\cdot \underbrace{\sum_{\pi' \in \mathcal{P}_{n-r}} \left(\prod_{l=1}^{k'-1} ((c + d) + dl) \prod_{j=1}^{k'} (1 - d)_{m_j-1} \right)}_{\text{Sum } S}.\end{aligned}$$

The term in parentheses is the numerator of the EPPF of a process $\text{PY}(d, c + d)$ defined on $n - r$ parameters and evaluated at partition π' . Since the sum S must equal the normalizing constant (the EPPF denominator) of the reduced process,

$$S = ((c + d) + 1)_{n-r-1}.$$

Substituting S in the preceding expression and using $(c + d)((c + d) + 1)_{n-r-1} = (c + d)_{n-r}$ gives

$$\mathbb{E}[K_{n,r}] = \binom{n}{r} \frac{(1 - d)_{r-1} (c + d)_{n-r}}{(c + 1)_{n-1}}.$$

□

Examining the asymptotic behaviour as $n \rightarrow \infty$ yields an expression that highlights the dependence on sample size. Using properties of the Gamma function,

$$\mathbb{E}[K_{n,r}] \sim \frac{\Gamma(c + 1)}{\Gamma(c + d)} \frac{\Gamma(r - d)}{r! \Gamma(1 - d)} n^d.$$

This asymptotic result is crucial to the characterization of the model. The expected number of clusters of size r diverges according to a power law n^d , analogous to the total expected number of clusters $\mathbb{E}[K_n]$.

This implies that, in the Pitman-Yor process, the number of small clusters continues to increase as the number of observations grows, confirming the model's ability to generate a partition with many distinct small groups.

In contrast, setting $d = 0$ (the Dirichlet process case) reduces the asymptotic expression to a constant:

$$\mathbb{E}[K_{n,r}] \longrightarrow \frac{c}{r} \quad (n \rightarrow \infty),$$

indicating that, in the Dirichlet process, the expected number of clusters of size r stabilizes at a finite value.

For a sample of n observations, the probability that a cluster selected uniformly from all generated clusters has size r is defined as

$$P(\text{size} = r) = \frac{\mathbb{E}[K_{n,r}]}{\mathbb{E}[K_n]}.$$

Translation note. This ratio-of-expectations definition is retained from the original. It describes pooling clusters over realizations; selecting a realization first and then choosing one of its clusters uniformly instead gives $\mathbb{E}[K_{n,r}/K_n]$.

In the Pitman-Yor process, the asymptotic limit stabilizes at a nonzero form:

$$\lim_{n \rightarrow \infty} P(\text{size} = r) = \frac{d\Gamma(r-d)}{\Gamma(1-d)r!} \sim \frac{d}{\Gamma(1-d)} r^{-(1+d)} \quad (r \rightarrow \infty),$$

indicating power-law decay in r . Thus, even with infinitely many observations, a positive and constant probability of observing clusters of specific sizes remains.

Figure 6 validates the theoretical results empirically: the decreasing linear pattern on a logarithmic scale, evident in both the simulations and the asymptotic theoretical curves, is the characteristic graphical representation of a power law whose slope varies with d . In particular, larger values of d make the slope steeper, increasing the probability of small clusters and decreasing that of large clusters.

In contrast, in the Dirichlet process this probability vanishes asymptotically. Because the expected number of clusters of size r converges to a constant (c/r), while the expected total number of clusters grows logarithmically ($c \log n$), their ratio tends to zero:

$$P(\text{size} = r) \sim \frac{1}{r \log n} \xrightarrow{n \rightarrow \infty} 0.$$

This indicates that, for sufficiently large n , the probability mass spreads over an infinite tail of new clusters, making the probability of observing a cluster of any particular finite size negligible.

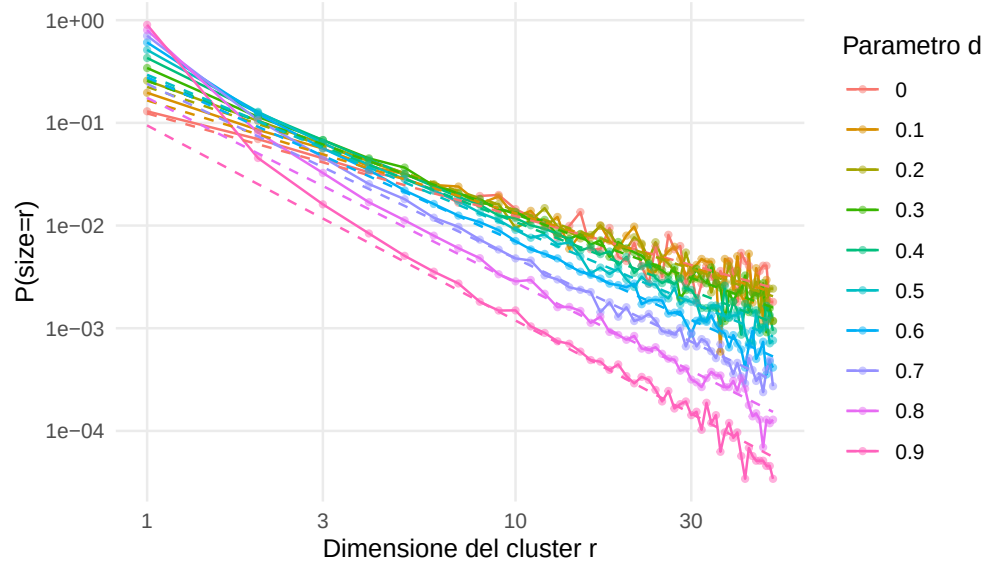


Figure 6: Distribution of $P(\text{size} = r)$ in the process $\text{PY}(c = 1, d)$ for varying d : comparison between simulation (solid lines) and asymptotic theoretical curves (dashed lines) on a logarithmic scale ($R = 300$ replications).

Original figure labels: “Dimensione del cluster r ”: cluster size r ; “Parametro d ”: parameter d .

4 Marginal algorithm for the PY mixture model

Consider a Pitman-Yor mixture model defined hierarchically as in Definition 2.2. The full conditional distribution of the latent variable θ_i , given the remaining parameters $\boldsymbol{\theta}_{-i}$ and observations $\mathbf{X}$, is

$$\begin{aligned}
 p(\theta_i \in dt \mid \boldsymbol{\theta}_{-i}, \mathbf{X}) &\propto \frac{c + k_{-i}d}{c + n - 1} \int K(x_i; \theta) g_0(\theta) d\theta \\
 &\quad \cdot \frac{K(x_i; t) g_0(t)}{\int K(x_i; \theta) g_0(\theta) d\theta} dt \\
 &\quad + \sum_{j=1}^{k_{-i}} \frac{n_{j,-i} - d}{c + n - 1} K(x_i; \theta_{j,-i}^*) \delta_{\theta_{j,-i}^*}(dt) \\
 &= \pi_0 \frac{K(x_i; t) g_0(t)}{\int K(x_i; \theta) g_0(\theta) d\theta} dt + \sum_{j=1}^{k_{-i}} \pi_j \delta_{\theta_{j,-i}^*}(dt) \\
 &= \pi_0 g(t; x_i) dt + \sum_{j=1}^{k_{-i}} \pi_j \delta_{\theta_{j,-i}^*}(dt).
 \end{aligned}$$

This full conditional was used to implement a Gibbs sampling algorithm for a PY($c = 1, d$) mixture with base measure $N(0, 1)$, univariate Gaussian kernel $N(\theta, \sigma^2)$, and parametric prior $\sigma^2 \sim \text{InvGamma}(2, 1)$.

The analysis was carried out on the `galaxies` dataset for ten different values of the parameter d .

Figure 7 shows substantial overlap among the curves, suggesting that the density estimate is robust to the considered variations in d .

A point estimate of the partition was obtained by minimizing the quadratic loss based on the similarity matrix, following the approach proposed by Dahl. The graphical analysis in Figure 8 shows marked sensitivity of the cluster structure to d . In agreement with the properties of the Pitman-Yor process, increasing d produces a substantial rise in the estimated number of groups. For large values of the parameter, this tendency leads to excessive fragmentation, with many clusters whose components often overlap spatially.

Density and cluster estimation were repeated for ten different values of c , as shown in Figures 9 and 10. As before, the density estimate remains almost unchanged as c varies, whereas a small increase in c causes many more clusters to be generated.

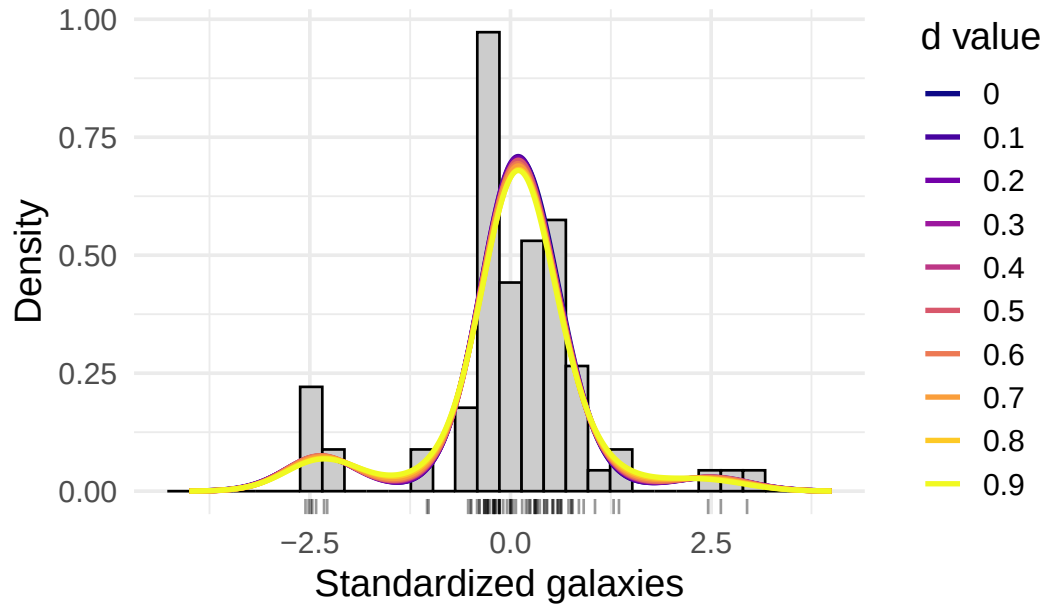


Figure 7: Density estimate for the standardized `galaxies` dataset obtained using a $PY(c = 1, d)$ mixture model for ten values of d , with 1000 iterations of the Gibbs sampling algorithm.

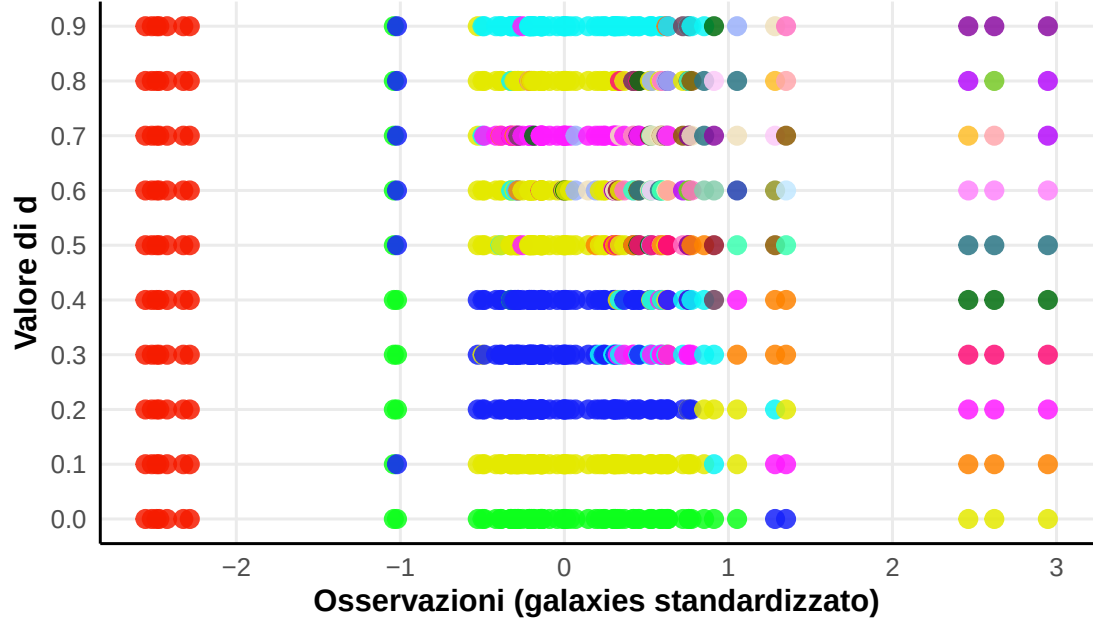


Figure 8: Estimated clusters for the standardized `galaxies` dataset obtained using a $PY(c = 1, d)$ mixture model for ten values of d , with 1000 iterations of the Gibbs sampling algorithm. The ideal partition was obtained by minimizing quadratic loss with respect to the similarity matrix, following Dahl’s approach.

Original figure labels: “Osservazioni (galaxies standardizzato)”: observations (standardized galaxies);
“Valore di d ”: value of d .

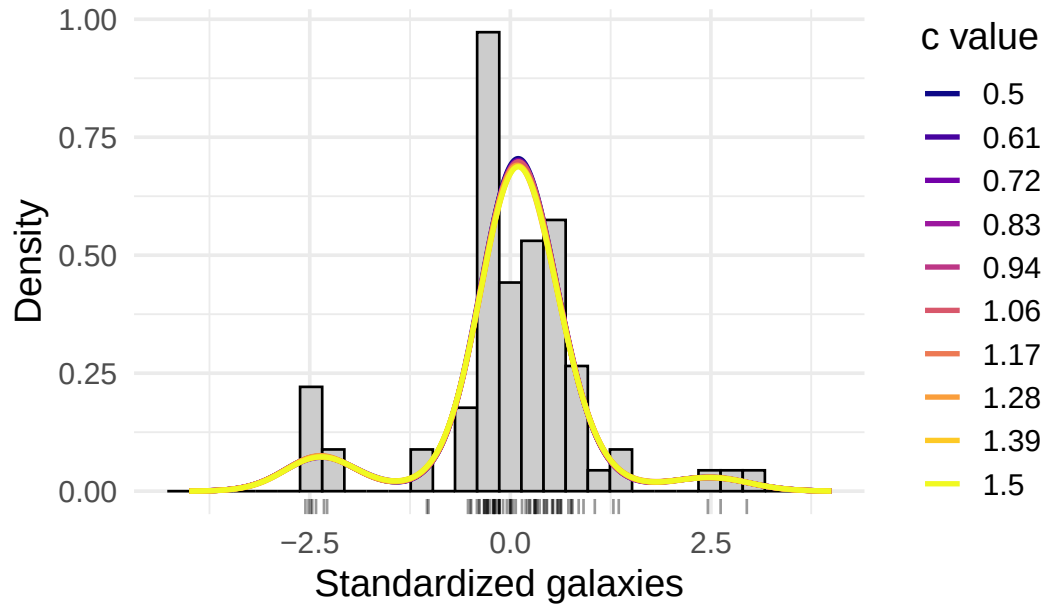


Figure 9: Density estimate for the standardized `galaxies` dataset obtained using a $PY(c, d = 0.3)$ mixture model for ten values of c , with 1000 iterations of the Gibbs sampling algorithm.

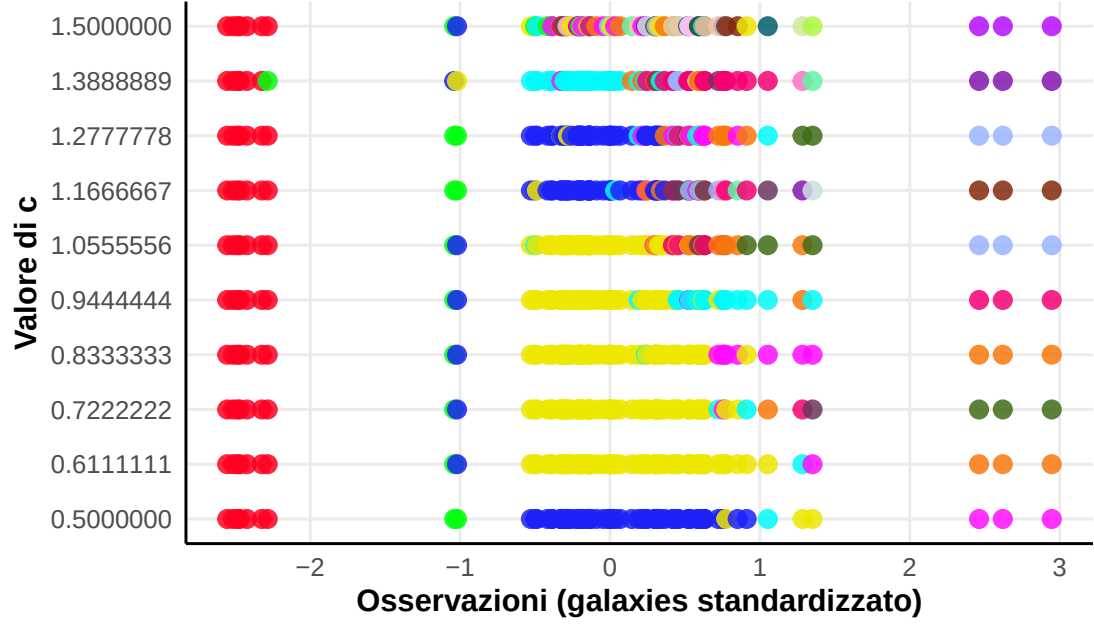


Figure 10: Estimated clusters for the standardized `galaxies` dataset obtained using a $PY(c, d = 0.3)$ mixture model for ten values of c , with 1000 iterations of the Gibbs sampling algorithm. The ideal partition was obtained by minimizing quadratic loss with respect to the similarity matrix, following Dahl's approach.

Original figure labels: "Osservazioni (galaxies standardizzato)": observations (standardized galaxies);
"Valore di c ": value of c .